ATC FlowLens

Explainable runway safety and air traffic control decision-support dashboard.

Helping air traffic teams see runway risk before it becomes a runway incident.

Team: Agentic Vibing

Focus: Advanced Technology and Innovation

Positioning: ATC support and training layer



SLIDE 2 - PROBLEM: RUNWAY OPERATIONS ARE TIME-CRITICAL

Controllers must make fast decisions while monitoring many moving parts at once.

The problem is not simply “many aircraft.” ATC workload rises when traffic, voice communication, weather, runway crossing, vehicle movement, and coordination happen at the same time.

What controllers must monitor

Aircraft approaching, landing, departing, or taxiing; active runway status; airport vehicles; runway crossing requests; weather and visibility; radio communication; readbacks; flight strips; and coordination across ATC units.

Why workload becomes risky

Critical information may be spread across different displays, voice communication, human memory, and procedure assumptions. When the time-to-conflict is short, even a small delay in recognizing the full picture can matter.

The runway conflict gap

A landing or departing aircraft can conflict with a vehicle if clearance status, vehicle movement, and aircraft timing are not interpreted together. ATC FlowLens focuses on that exact intersection.

The opportunity is not to replace controllers, but to give them a clearer operational lens when workload is already high.

SLIDE 3 - EVIDENCE: RUNWAY CONFLICT IS A REAL SAFETY ISSUE

Runway conflicts and workload-related incidents have happened in real operations.

345

runway incursions across FAA Core 30 airports in FY2024, down from 362 in FY2023.

391

average annual incursions across Core 30 airports before the pandemic, FY2017-FY2019.

12

Core 30 airports exceeded their pre-pandemic incursion levels in FY2024.

DFW/BOS/ORD

had the highest Core 30 incursion counts in FY2024.

Case	What happened	What it proves for FlowLens
LATAM Peru Flight 2213, 2022	An A320neo collided with a fire truck on the runway during takeoff roll in Lima.	Aircraft-vehicle runway conflict can form during emergency vehicle activity and coordination gaps.
Austin-Bergstrom, 2023	A departure aircraft was cleared while another aircraft was on final approach in dense fog.	Low visibility and runway status awareness can reduce the safety margin quickly.
LaGuardia, 2026 preliminary	NTSB opened an investigation after a CRJ-900 collided with an ARFF vehicle on runway 4.	Vehicle tracking, last-second clearance, transponder coverage, and workload are not abstract concerns.

Sources: **FAA Air Traffic by the Numbers FY2024**, **NTSB DCA26MA161**, **BEA LATAM 2213 event page**, **AP/NTSB Austin coverage**.

ATC already has important technologies, but the challenge is connecting them into one clear operational picture.

Existing ATC / ATM technologies

Modern ATC/ATM uses radar, ADS-B, MLAT, surface surveillance, electronic flight strips, A-SMGCS, RIMCAS/RMCA, STCA/MTCD, CPDLC/Data Comm, AMAN/DMAN, TFDM, and SWIM. These systems are important, but they are not equally available everywhere.

Coverage and fragmentation reality

Advanced surface tools such as ASDE-X, ASSC, A-SMGCS, and RIMCAS are valuable but can be expensive and complex. Voice communication also remains common, especially in tower and ground operations, so clearance state and runway timing can still require manual interpretation.

Where ATC FlowLens fits

Existing data

Aircraft, vehicles, runway, clearance, weather, communication logs.

→

FlowLens layer

Time-to-conflict, workload scoring, event explanation, replay.

→

Human decision

Controller remains in command. The tool supports awareness and training.

Sources: **ICAO Global Runway Safety Action Plan**, **MITRE surface safety prototypes**, **SKYbrary RIMCAS**, **FAA AIM ASDE-X/ASSC**.

SLIDE 5 - SOLUTION: ATC FLOWLENS

ATC FlowLens brings runway movement, clearance status, vehicle activity, communication, and workload risk into one dashboard.

The product does not automate ATC. It connects operational signals into a single timeline so teams can detect, explain, and replay runway risk.

Runway occupancy model

Tracks whether a runway is clear, occupied, being crossed, used for takeoff, or receiving an inbound landing aircraft.

Clearance state engine

Maintains operational states such as cleared to land, cleared for takeoff, hold short, line up and wait, cleared to cross, and go around.

Time-to-conflict

Compares estimated aircraft touchdown or runway entry time with estimated vehicle runway clear time.

Communication timeline

Organizes tower, pilot, and vehicle messages into an event history that can later support speech-to-text or readback mismatch detection.

Workload indicator

Scores workload using traffic count, radio calls, active clearances, vehicle movements, weather, runway changes, and active alerts.

Replay and explanation

Turns a risk event into a training object: what happened, when it happened, why the alert appeared, and what action category should be considered.

SLIDE 6 - HOW IT WORKS

The system converts simulated operational data into clear runway risk insight.

1

Aircraft inbound

BAW123 is 1.8 NM final, speed 145 kt, cleared to land RWY 27.

2

Vehicle crossing

ARFF-2 is cleared or starts to cross RWY 27 during an active runway operation.

3

Prediction

The engine estimates touchdown in 42 seconds and vehicle clear time in 58 seconds.

4

Conflict window

Vehicle clear time is later than aircraft touchdown time. Predicted overlap: 16 seconds.

5

Explainable output

The dashboard shows alert severity, reason, relevant objects, workload state, and replay timeline.

Example alert

HIGH RUNWAY CONFLICT RISK. BAW123 is cleared to land RWY 27. ARFF-2 is cleared to cross RWY 27. Vehicle clear time is later than aircraft touchdown time.

Why this matters in a pitch

The demo has visible inputs, processing, and outputs: aircraft/vehicle data, clearance state, time-to-conflict, alert explanation, workload state, and replay timeline.

SLIDE 7 - MVP SCOPE AND DEMO PLAN

The MVP is feasible for a hackathon because it uses simulation, rule-based logic, and a focused runway scenario.

Hackathon MVP

One simulated airport layout with runway and taxiway, three to five aircraft, one to two airport vehicles, clearance and communication logs, weather or visibility status, rule-based conflict detection, time-to-conflict alerting, workload scoring, and replay mode.

Why this is feasible

It avoids live ATC data, regulatory integration, and certification dependency. The team can still demonstrate a credible safety concept using realistic scenarios and transparent logic.

Why this is valuable

Training centers and airport safety teams need explainable scenarios, not only raw alarms. FlowLens can become a reusable safety learning environment.

Layer	What we build now	What it can become
Data	Scenario JSON and simulated positions.	ADS-B, airport operation data, weather feed, training simulator logs.
Logic	Rule-based time-to-conflict and workload score.	Richer prediction, scenario generator, human factors model.
Interface	Runway map, alert reason, event replay.	Training lab, safety analytics, research collaboration with ANSP/aviation school.

The first adoption path is training, simulation, and safety awareness before real operational integration.

Problem owners

Airport operators, Air Navigation Service Providers, ATC training centers, aviation academies, airport safety teams, and airport rescue and firefighting units.

Adoption path

1. Training and simulation tool. 2. Safety briefing and scenario replay product. 3. Airport operation support prototype. 4. Research collaboration. 5. Future certified decision-support exploration.

Training

Better runway conflict scenarios and controller workload awareness.

Review

Faster explanation of complex incident timelines and near-miss patterns.

Readiness

Safer coordination between tower, ground, emergency units, and airport operations.

Expansion

Speech-to-text, readback mismatch, richer workload analytics, and simulator integration.

A simulation-first product is more fundable and buildable than a premature claim to replace certified ATC infrastructure.

SLIDE 9 - EXPECTED IMPACT AND BUILD INDONESIA ALIGNMENT

ATC FlowLens aligns with BUILD Indonesia by applying advanced technology to a real aerospace safety challenge.

Impact area	Expected value	BUILD Indonesia alignment
Runway safety	Helps users recognize runway conflict risk earlier by showing aircraft, vehicle, clearance, and timing in one view.	Aerospace & Defense: supports airport operational readiness and critical aviation safety.
Training and review	Makes training scenarios easier to explain and replay, especially for aircraft-vehicle conflict and communication sequence.	Advanced Technology and Innovation: uses digital decision support, simulated tracking, and explainable alerting.
Workload awareness	Gives a structured workload indicator based on traffic, radio calls, weather, vehicle movement, clearance load, and alerts.	Social Impact: safer operations protect passengers, crews, ground staff, and emergency units.
Operational flow	Future versions may reduce avoidable go-arounds, taxi delay, holding, and unnecessary fuel burn.	Sustainability: better runway flow can reduce waste in future operational use.

Primary focus area

Advanced Technology and Innovation through decision support, prediction, workload scoring, and explainable runway safety visualization.

Secondary relevance

Social Impact and Sustainability through safer runway operations, better coordination, and future reduction of avoidable fuel waste.

Why it is finalist-worthy

The idea is specific, evidence-backed, feasible as an MVP, and positioned carefully for a safety-critical aviation domain.

ATC FlowLens starts as a feasible hackathon MVP and can grow into a broader aviation safety training platform.

Phase 1: Hackathon MVP

Simulated runway map, aircraft and vehicle movement, clearance log, time-to-conflict alert, workload indicator, and replay timeline.

Phase 2: Enhanced prototype

ATC-style speech-to-text, readback mismatch detection, scenario generator, multi-runway support, and richer workload model.

Phase 3: Future vision

Integration with airport operation datasets, tower-ground training simulator, safety analytics dashboard, and research pathway toward certified support.

Why BUILD Indonesia: mentorship, validation, and aerospace expertise can turn this from a strong prototype into a credible aviation safety product direction.

Final statement

ATC FlowLens is not trying to automate the controller. It gives controllers, trainers, and safety teams a clearer lens on runway flow, workload, and conflict risk when every second matters.